

Does Leverage Cause Corporate Financial Distress?

A causal machine learning analysis of U.S. public firms using Compustat financial statements

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1. Motivation and economic relevance

Why the distinction between prediction and causality matters for credit risk decisions

ECONOMIC PROBLEM

Financial distress is costly before default appears

- Firm value loss, layoffs and spillovers to banks and credit markets
- Risk managers need early warning models using observable accounting data
- Altman Z-Score is still a common benchmark, but it is static and hand-designed



CORE RELEVANCE

Leverage can mean two very different things

Cause

Debt pressure pushes firms into distress

or

Symptom

Weak firms borrow because operations already deteriorate

Managerial implication

If leverage is causal, debt regulation and covenant limits matter. If it is endogenous, purely leverage-based rules may misdiagnose risk.

2. Data foundation

EDA highlights from the final Compustat firm-year panel used for modelling

66,580

final firm-year observations

8,508

unique firms

2000–2021

modelling sample

8.7%

new distress transition rate

2017+

out-of-time test period

EDA view: sample construction

Cleaning turns the raw WRDS pull into the final modelling panel.

Raw Compustat pull

220,729 firm-years · 25,056 firms

After industry / duplicate filters

137,470 firm-years · 16,107 firms

Final modelling sample

66,580 firm-years · 8,508 firms

Handled before modelling: financials/utilities, impossible values, stub periods, already-distressed years.

Target and time split

The label is rare and evaluated out of time.

Distress in t+1



8.7% new distress transitions

No-leakage split



49,280

Train

5,076

Validation

12,224

Test

Variable design

One row is one firm-year; features are lagged relative to next-year distress.

Target

distress in t+1

negative equity or interest coverage < 1

Treatment

long-term leverage

long-term debt / total assets

Controls

accounting ratios + size_log + sich

profitability, liquidity, efficiency, solvency

Benchmark

Altman Z-Score

classical ratio-based benchmark feature

EDA takeaway

The dataset is clean, panel-based and rare-event by design; the time split and lagged accounting ratios support both prediction and causal analysis.

3. Methodology design

The notebook combines causal identification, supervised prediction and unsupervised segmentation on the same firm-year dataset

Three analytical blocks

A

IDENTIFICATION

Causal inference

Backdoor adjustment with DoWhy DAG

Treatment: long-term leverage

Outcome: next-year distress

Controls: profitability, size, liquidity, efficiency and historical industry (sic)

Validation: naive vs. adjusted effect + placebo refuter

B

PREDICTION

Supervised learning

Compare linear and nonlinear classifiers

Models: Lasso, Logistic, SVM, Random Forest, Neural Net

Time split: train 2000–2014; validation 2015–2016; test 2017–2021

Metrics: AUC, F1, accuracy, ROC and precision-recall curves

C

SEGMENTATION

Unsupervised learning

PCA visualization + KMeans clustering

Uses financial ratios only, without distress labels

PCA reduces standardized ratios to two dimensions

Clusters interpreted using ratio profile, Altman Z and empirical distress rate

Purpose: identify financial health profiles beyond prediction

4. Results

Three method blocks jointly evaluate causality, prediction and financial-risk segmentation

A. CAUSAL INFERENCE

ATE = +0.0906

DoWhy backdoor estimate

- Naive leverage coefficient: -0.400
- Adjusted coefficient: +0.829
- Placebo new effect: 0.0024

Leverage turns positive after controlling for confounding.

B. SUPERVISED LEARNING

Best: Random Forest

AUC = 0.811

- NN 0.806 · SVM 0.799 · Logistic 0.748
- RF: Precision 0.267 · Recall 0.602 · F1 0.370
- Key predictors: size_log, profit_margin, retained_ratio

Nonlinear models discriminate better; precision remains limited.

C. UNSUPERVISED LEARNING

PCA + KMeans

k = 3 clusters

- C0: leverage 0.024 · distress 9.6%
- C1: leverage 0.316 · distress 5.9%
- C2: leverage 0.134 · distress 11.5%

Risk is not monotonic in leverage; overall profile matters.

Synthesis

Causal

Leverage is causally relevant after adjustment.

Predictive

Broader liabilities and profitability carry strong signal.

Overall message

Distress prediction should use financial health, not leverage alone.

5. Limitations and synthesis

Discussion and conclusion: what the three methods jointly imply, and where the analysis remains limited

SYNTHESIS ACROSS METHODS

The three method blocks point to the same core message:

Causal

After adjustment, leverage shows a positive causal effect on distress risk.

Supervised

Random Forest performs best out-of-time; broader liabilities and profitability carry strong predictive signal.

Unsupervised

Clusters show leverage alone does not determine risk; overall financial health matters.

MAIN LIMITATIONS

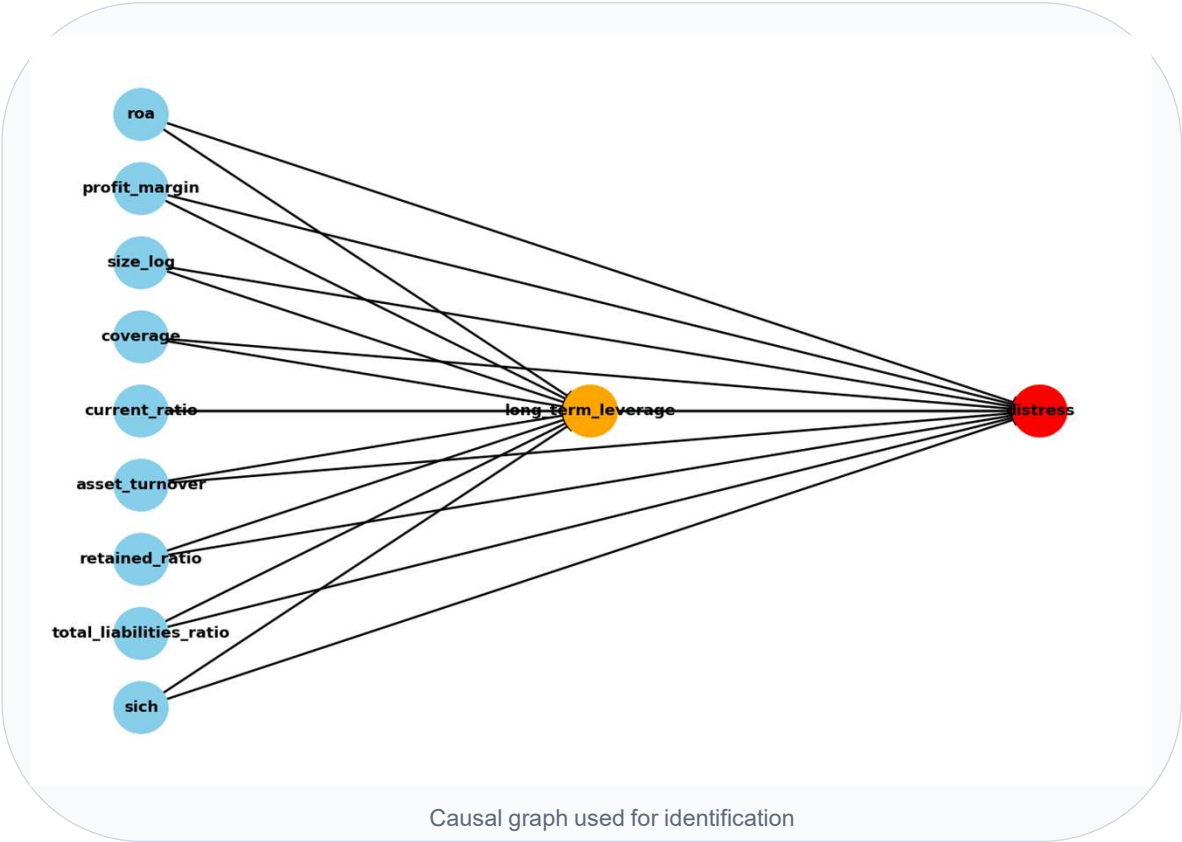
- Distress is a constructed proxy, not a verified bankruptcy filing.
- Extreme asset jumps were kept; excluding them would be a useful robustness check.
- SVM was trained on an 8,000-row subsample for computational reasons.

Best next step: validate the label using bankruptcy filings or CRSP delisting codes.

Bottom line: leverage is causally relevant, but practical distress prediction needs broader financial-health indicators, not long-term debt alone.

Backup A. Causal inference details

Identification graph and key estimates supporting the main results slide



KEY ESTIMATES

Naive logistic coefficient	-0.400 (p < 0.001)
Adjusted coefficient	+0.829 (p < 0.001)
Odds ratio for +0.1 leverage	1.086
DoWhy backdoor ATE	+0.0906
Placebo refuter new effect	0.0024

Backup B. Supervised learning details

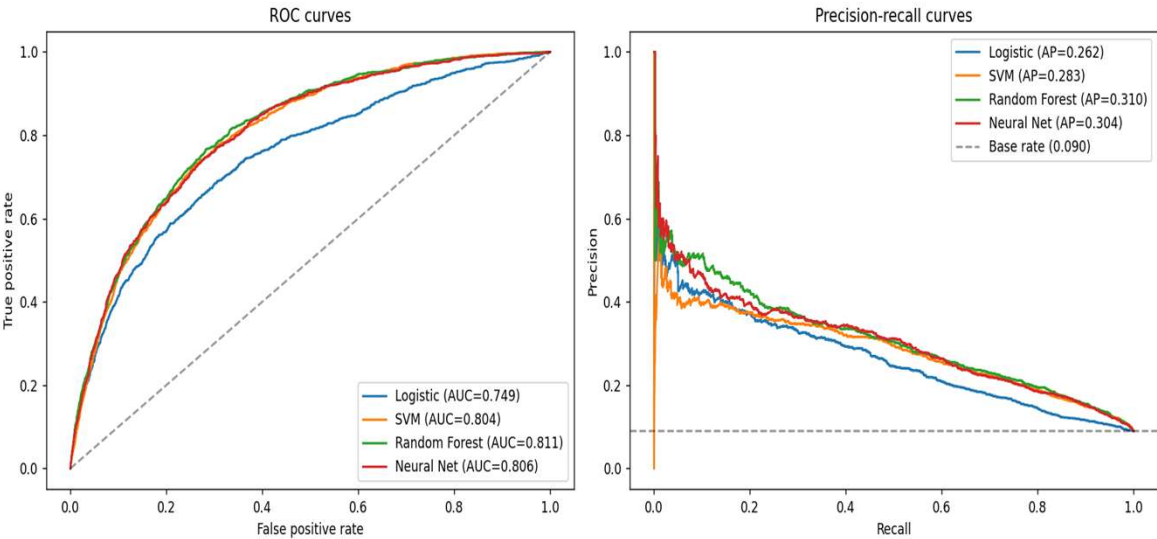
Out-of-time model comparison, ROC/PR performance and Random Forest feature importance

Train
49,280 (2000–2014)

Validation
5,076 (2015–2016)

Test
12,224 (2017–2021)

One-year transition to distress
8.7%

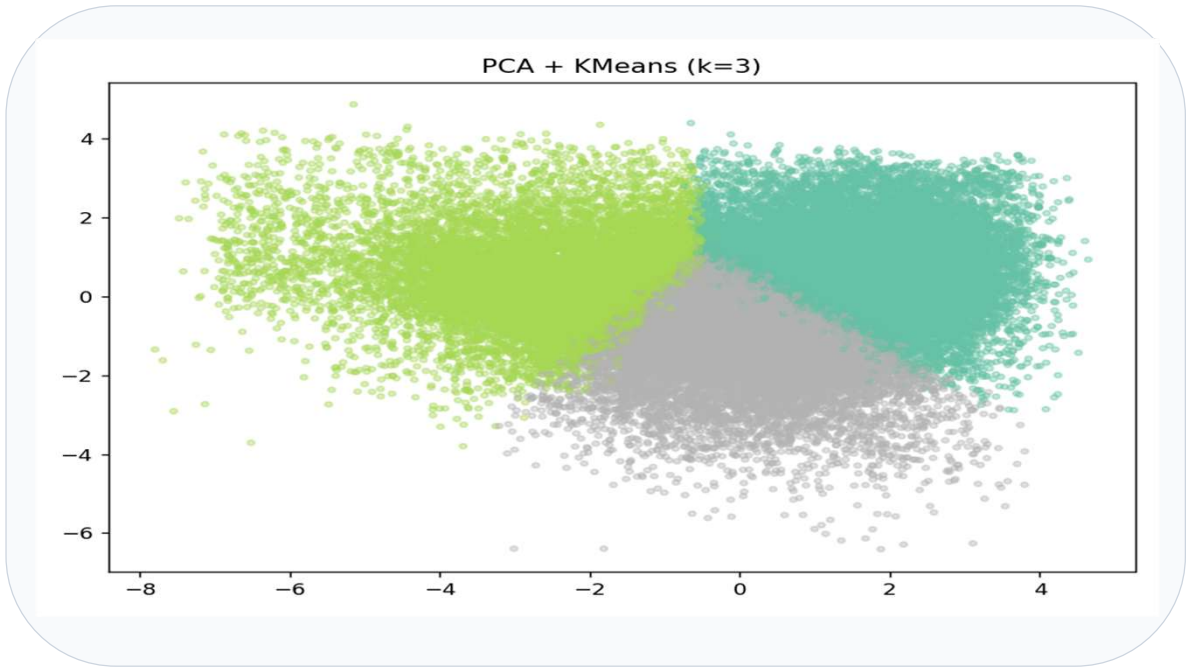


TEST-SET METRICS

Model	AUC	Acc.	F1
Random Forest	0.811	0.816	0.370
Neural Net	0.806	0.724	0.323
SVM	0.799	0.797	0.349
Logistic	0.748	0.780	0.318

Backup C. Unsupervised learning details

PCA visualization and cluster profiles from KMeans segmentation (k = 3)



CLUSTER PROFILES

Cluster	LT lev.	TL ratio	Altman Z	Distress
0	0.024	0.242	6.030	9.6%
1	0.316	0.624	1.718	5.9%
2	0.134	0.518	2.738	11.5%